

KAGE Fitness Studio ERP — Detailed Data Dictionary

A field-level business data dictionary showing what data should be captured for each ERP entity, why it is needed, and what validation or business rule applies.

Project Type Independent Business Analyst Portfolio Case Study	Domain Fitness Studio / ERP / Gym Operations
Prepared by Adina Mohammed	Date May 2026
Portfolio Boundary Note This is a fictional BA portfolio artefact inspired by real-world gym operations. It is not a live client project, production ERP system, or final technical database design. The purpose is to demonstrate Business Analyst thinking through structured data documentation.	

DOCUMENT PURPOSE

Why This Document Exists

This data dictionary explains what data should be captured for each ERP entity, why it is needed, and what business rule or validation applies.

It is designed to be recruiter-friendly, simple to read, and easy to convert into a PDF for portfolio presentation.

DATA DICTIONARY OVERVIEW

Document Scope

The KAGE ERP data model covers memberships, payments, PT sessions, freelancer billing, equipment maintenance, and reporting.

Each entity has been documented at field level to show how business information supports daily operations.

Simple BA data types Text, Number, Date, Time, Currency, Dropdown, Boolean, Auto-generated ID, Calculated, and Reporting value.

Member Entity

FIELD NAME	DATA TYPE	DESCRIPTION	EXAMPLE VALUE	MANDATORY	VALIDATION / BUSINESS RULE
MemberID	Auto-generated ID	Unique identifier for each member.	MBR-10021	Yes	Must be system-generated and unique.
FullName	Text	Member’s full name.	Rahul Kumar	Yes	Must not be blank.
Phone	Text	Member contact number.	9876543210	Yes	Must be unique; duplicate phone not allowed.
Email	Text	Member email address.	rahul@gmail.com	No	If entered, must be unique; duplicate email not allowed.
MemberType	Dropdown	Category of member in the system.	Direct Member	Yes	Allowed values: Direct Member, Freelancer, Freelancer Client.
JoinDate	Date	Date the member joined.	2026-05-10	Yes	Cannot be a future date.
Status	Dropdown	Current membership state.	Active	Yes	Allowed values: Active, Expired, On Hold, Inactive.
PlanID	Auto-generated ID	Links the member to a membership plan.	PLN-001	Yes	Must reference a valid MembershipPlan record.

MembershipPlan Entity

FIELD NAME	DATA TYPE	DESCRIPTION	EXAMPLE VALUE	MANDATORY	VALIDATION / BUSINESS RULE
PlanID	Auto-generated ID	Unique identifier for the plan.	PLN-001	Yes	Must be system-generated and unique.
PlanName	Text	Name of the membership plan.	Monthly Plan	Yes	Must not be blank.
Duration	Number	Plan duration in days or months.	30	Yes	Must be greater than zero.
BaseFee	Currency	Standard fee for the plan.	2500	Yes	Must be greater than zero.
IsNegotiable	Dropdown	Indicates whether the fee can be negotiated.	No	Yes	Allowed values: Yes, No.
Status	Dropdown	Availability of the plan.	Active	Yes	Allowed values: Active, Inactive.

Payment Entity

FIELD NAME	DATA TYPE	DESCRIPTION	EXAMPLE VALUE	MANDAT ORY	VALIDATION / BUSINESS RULE
PaymentID	Auto-generated ID	Unique identifier for each payment.	PAY-50012	Yes	Must be system-generated and unique.
MemberID	Auto-generated ID	Member who made the payment.	MBR-10021	Yes	Must reference a valid Member.
PaymentPurpose	Dropdown	Reason for the payment.	Membership	Yes	Allowed values: Membership, PT Package, Renewal, Freelancer Fee, Maintenance Expense, Other.
Amount	Currency	Payment amount.	2500	Yes	Must not be blank, zero, or negative.
PaymentMode	Dropdown	How the payment was made.	UPI	Yes	Allowed values: Cash, UPI, Card, Bank Transfer.
PaymentStatus	Dropdown	Current status of the payment.	Paid	Yes	Allowed values: Paid, Partial, Pending, Overdue, Refunded.
InvoiceNumber	Text	Invoice reference number.	INV-2026-041	Yes	Must be unique.
InvoiceSharedStatus	Dropdown	Tracks whether invoice was shared.	Sent	Yes	Allowed values: Not Shared, Sent, Downloaded.
PaymentDate	Date	Date of payment collection.	2026-05-12	Yes	Cannot be a future date.
CollectedBy	Text	User or staff member who collected the payment.	Front Desk	Yes	Must not be blank.

Trainer Entity

FIELD NAME	DATA TYPE	DESCRIPTION	EXAMPLE VALUE	MANDAT ORY	VALIDATION / BUSINESS RULE
TrainerID	Auto-generated ID	Unique trainer identifier.	TRN-003	Yes	Must be system-generated and unique.
FullName	Text	Trainer’s name.	Amit Sharma	Yes	Must not be blank.
Specialisation	Dropdown	Primary training focus.	Strength	Yes	Allowed values include Weight Loss, Strength, Rehab Support, General Fitness, Body Transformation.
ShiftStatus	Dropdown	Current trainer availability.	Available	Yes	Allowed values: Available, Busy, On Break, Absent.
CommissionRate	Number	Agreed commission percentage or value.	12	Yes	Must be a valid percentage or agreed rate.
Status	Dropdown	Employment or active state.	Active	Yes	Allowed values: Active, Inactive.

PTPackage Entity

FIELD NAME	DATA TYPE	DESCRIPTION	EXAMPLE VALUE	MANDATORY	VALIDATION / BUSINESS RULE
PackageID	Auto-generated ID	Unique PT package identifier.	PTP-20011	Yes	Must be system-generated and unique.
MemberID	Auto-generated ID	Member who purchased the package.	MBR-10021	Yes	Must reference a valid Member.
TotalSessions	Number	Total number of sessions in the package.	12	Yes	Must be greater than zero.
CompletedSessions	Number	Number of completed sessions.	4	Yes	Must not exceed TotalSessions.
RemainingSessions	Calculated	Balance left in the package.	8	Yes	Must equal TotalSessions - CompletedSessions.
PackageStatus	Dropdown	Current package state.	Active	Yes	Allowed values: Active, Completed, Cancelled, On Hold.

PTSession Entity

FIELD NAME	DATA TYPE	DESCRIPTION	EXAMPLE VALUE	MANDATORY	VALIDATION / BUSINESS RULE
SessionID	Auto-generated ID	Unique session identifier.	SESS-90034	Yes	Must be system-generated and unique.
MemberID	Auto-generated ID	Member attending the session.	MBR-10021	Yes	Must reference a valid Member.
TrainerID	Auto-generated ID	Trainer conducting the session.	TRN-003	Yes	Must reference a valid Trainer.
PackageID	Auto-generated ID	PT package linked to the session.	PTP-20011	Yes	Must reference a valid PTPackage.
SessionDate	Date	Date of the PT session.	2026-05-14	Yes	Cannot be a future date if marked completed.
StartTime	Time	Session start time.	07:00 AM	Yes	Must be a valid time.
EndTime	Time	Session end time.	08:00 AM	Yes	Must be later than StartTime.
SessionStatus	Dropdown	Status of the PT session.	Completed	Yes	Allowed values: Scheduled, Completed, Cancelled, No Show, Rescheduled.
CommissionStatus	Dropdown	Commission state for the trainer.	Estimated	Yes	Allowed values: Not Eligible, Estimated, Approved, Paid.

FREELANCER

Freelancer Entity

FIELD NAME	DATA TYPE	DESCRIPTION	EXAMPLE VALUE	MANDATORY	VALIDATION / BUSINESS RULE
FreelancerID	Auto-generated ID	Unique freelancer identifier.	FRL-001	Yes	Must be system-generated and unique.
MemberID	Auto-generated ID	Links freelancer to the Member record.	MBR-20005	Yes	Freelancer must also exist as a Member with MemberType = Freelancer.
MonthlyAccessFee	Currency	Freelancer's own monthly access fee.	5000	Yes	Must be greater than zero.
ActiveClientCount	Number	Number of active freelancer clients.	4	Yes	Must be based on linked active FreelancerClient records.
AccessFlagStatus	Dropdown	Access review state.	Clear	Yes	Allowed values: Clear, Warning, Gate Flag Active.
PaymentStatus	Dropdown	Monthly fee payment state.	Paid	Yes	Allowed values: Paid, Pending, Overdue.

FreelancerClient Entity

FIELD NAME	DATA TYPE	DESCRIPTION	EXAMPLE VALUE	MANDATORY	VALIDATION / BUSINESS RULE
FreelancerClientID	Auto-generated ID	Unique freelancer client record.	FCL-30018	Yes	Must be system-generated and unique.
MemberID	Auto-generated ID	Member linked as a freelancer client.	MBR-30018	Yes	FreelancerClient must also exist as a Member with MemberType = Freelancer Client.
FreelancerID	Auto-generated ID	Freelancer to whom the client is linked.	FRL-001	Yes	Each FreelancerClient must be linked to one Freelancer.
PackageType	Dropdown	Client package type.	Monthly	Yes	Allowed values: Monthly, Quarterly, Annual, Custom.
ActiveStatus	Dropdown	Whether the link is active.	Active	Yes	Allowed values: Active, Inactive.
LinkedDate	Date	Date the client was linked.	2026-05-15	Yes	Cannot be a future date.

FreelancerBilling Entity

FIELD NAME	DATA TYPE	DESCRIPTION	EXAMPLE VALUE	MANDATORY	VALIDATION / BUSINESS RULE
BillingID	Auto-generated ID	Unique billing record identifier.	BIL-0044	Yes	Must be system-generated and unique.
FreelancerID	Auto-generated ID	Freelancer for whom billing is calculated.	FRL-001	Yes	Must reference a valid Freelancer.
BillingMonth	Date	Month for which billing is generated.	2026-05-01	Yes	Must represent the billing period.
OwnFee	Currency	Freelancer’s own monthly fee.	5000	Yes	Must be greater than zero.
ActiveClientCount	Number	Number of active clients for that month.	4	Yes	Must be based on active FreelancerClient records.
PerClientAccessFee	Currency	Fee charged per active client.	3000	Yes	Must be greater than or equal to zero.
MonthlyPayable	Calculated	Total monthly amount due.	17000	Yes	MonthlyPayable = OwnFee + (ActiveClientCount × PerClientAccessFee).
DueDate	Date	Payment due date.	2026-05-07	Yes	Must be within the billing cycle.
PaymentStatus	Dropdown	Current payment state.	Pending	Yes	Allowed values: Paid, Pending, Overdue, Disputed.

EQUIPMENT

Equipment Entity

FIELD NAME	DATA TYPE	DESCRIPTION	EXAMPLE VALUE	MANDATORY	VALIDATION / BUSINESS RULE
EquipmentID	Auto-generated ID	Unique equipment identifier.	EQP-010	Yes	Must be system-generated and unique.
EquipmentName	Text	Name of the equipment item.	Treadmill 02	Yes	Must not be blank.
Category	Dropdown	Equipment category.	Cardio	Yes	Must be a defined business category.
CurrentStatus	Dropdown	Current condition of the equipment.	Operational	Yes	Allowed values: Operational, Under Repair, Out of Service, Decommissioned.
PurchaseDate	Date	Date equipment was purchased.	2024-03-18	Yes	Cannot be a future date.
ReplacementReviewFlag	Boolean	Indicates whether replacement review is required.	Yes	Yes	Allowed values: Yes, No.

MaintenanceLog Entity

FIELD NAME	DATA TYPE	DESCRIPTION	EXAMPLE VALUE	MANDATORY	VALIDATION / BUSINESS RULE
MaintenanceID	Auto-generated ID	Unique maintenance record identifier.	MNT-7801	Yes	Must be system-generated and unique.
EquipmentID	Auto-generated ID	Equipment item linked to the repair.	EQP-010	Yes	Each maintenance log must be linked to one equipment item.
RepairDate	Date	Date of repair or service event.	2026-05-20	Yes	Cannot be a future date.
RepairCost	Currency	Cost of the repair.	1500	Yes	Must be zero or greater.
VendorName	Text	Repair vendor or service provider.	TechFix Services	No	If entered, should not be blank.
IssueNotes	Text	Notes about the repair issue.	Belt worn out	Yes	Must describe the issue clearly enough for business review.
ResolutionStatus	Dropdown	Current resolution status.	Resolved	Yes	Allowed values: Open, Under Review, Resolved, Pending Vendor, Escalated.

RevenueReport Entity

FIELD NAME	DATA TYPE	DESCRIPTION	EXAMPLE VALUE	MANDATORY	VALIDATION / BUSINESS RULE
ReportID	Auto-generated ID	Unique report reference.	RPT-2026-05	Yes	Must be system-generated and unique.
ReportPeriod	Dropdown	Reporting period used for analysis.	Monthly	Yes	Allowed values: Weekly, Monthly, Quarterly, Custom date range.
MembershipRevenue	Reporting value	Total membership income for the period.	85000	Yes	Summarised from Payment records.
PTRevenue	Reporting value	Total PT-related income for the period.	42000	Yes	Summarised from Payment / PTPackage records.
FreelancerRevenue	Reporting value	Total freelancer billing income.	24000	Yes	Summarised from FreelancerBilling.
MaintenanceCost	Reporting value	Total maintenance expense for the period.	6000	Yes	Summarised from MaintenanceLog.
NetRevenue	Calculated	Overall net business result.	145000	Yes	NetRevenue = MembershipRevenue + PTRevenue + FreelancerRevenue - MaintenanceCost - other recorded expenses.

VALIDATION SUMMARY

Key Validation Rules

AREA	VALIDATION RULE	WHY IT MATTERS
Member identity	Phone and email must not be duplicated.	Prevents duplicate member profiles and supports clean customer records.
Membership plan	BaseFee must be greater than zero.	Ensures plans are financially valid.
Payment capture	Amount cannot be blank, zero, or negative.	Protects billing accuracy and revenue reporting.
Invoice tracking	InvoiceNumber must be unique.	Avoids invoice duplication and confusion during audit or follow-up.
PT package balance	CompletedSessions cannot exceed TotalSessions.	Prevents invalid package usage and inaccurate balances.
Trainer scheduling	Trainer double-booking must be blocked unless shared/group PT is approved.	Maintains schedule control and avoids operational conflicts.
Freelancer billing	Billing must remain monthly and access-fee based.	Kept aligned to the business model and avoids incorrect session-based charging.
Maintenance control	Each maintenance log must link to one equipment item.	Supports traceability and equipment history review.
Replacement logic	Repeated repairs can trigger ReplacementReviewFlag.	Supports equipment lifecycle decisions.

AREA	VALIDATION RULE	WHY IT MATTERS
Reporting logic	RevenueReport must be summarised, not manually transacted.	Keeps reports business-focused and prevents double counting.

DROPDOWN SUMMARY

Dropdown and Status Values

FIELD	ALLOWED VALUES
MemberType	Direct Member, Freelancer, Freelancer Client
Status	Active, Expired, On Hold, Inactive
MembershipPlan.Status	Active, Inactive
MembershipPlan.IsNegotiable	Yes, No
PaymentMode	Cash, UPI, Card, Bank Transfer
PaymentStatus	Paid, Partial, Pending, Overdue, Refunded
InvoiceSharedStatus	Not Shared, Sent, Downloaded
PaymentPurpose	Membership, PT Package, Renewal, Freelancer Fee, Maintenance Expense, Other
ShiftStatus	Available, Busy, On Break, Absent
Trainer.Status	Active, Inactive
PTPackage.PackageStatus	Active, Completed, Cancelled, On Hold
PTSession.SessionStatus	Scheduled, Completed, Cancelled, No Show, Rescheduled
PTSession.CommissionStatus	Not Eligible, Estimated, Approved, Paid

FIELD	ALLOWED VALUES
Freelancer.AccessFlagStatus	Clear, Warning, Gate Flag Active
Freelancer.PaymentStatus	Paid, Pending, Overdue
FreelancerClient.PackageType	Monthly, Quarterly, Annual, Custom
FreelancerClient.ActiveStatus	Active, Inactive
FreelancerBilling.PaymentStatus	Paid, Pending, Overdue, Disputed
Equipment.CurrentStatus	Operational, Under Repair, Out of Service, Decommissioned
Equipment.ReplacementReviewFlag	Yes, No
MaintenanceLog.ResolutionStatus	Open, Under Review, Resolved, Pending Vendor, Escalated
RevenueReport.ReportPeriod	Weekly, Monthly, Quarterly, Custom date range

CALCULATED FIELDS

Calculated Fields Summary

CALCULATED FIELD	FORMULA / LOGIC	SOURCE ENTITIES
RemainingSessions	TotalSessions - CompletedSessions	PTPackage
MonthlyPayable	OwnFee + (ActiveClientCount × PerClientAccessFee)	FreelancerBilling
ActiveClientCount	Count of active linked FreelancerClient records	FreelancerClient, Freelancer
NetRevenue	MembershipRevenue + PTRevenue + FreelancerRevenue - MaintenanceCost - other recorded expenses	RevenueReport, Payment, PTPackage, FreelancerBilling, MaintenanceLog
ReplacementReviewFlag	Set to Yes when repeated repairs cross the defined review threshold	Equipment, MaintenanceLog